HARDWARE REQUIREMENTS:

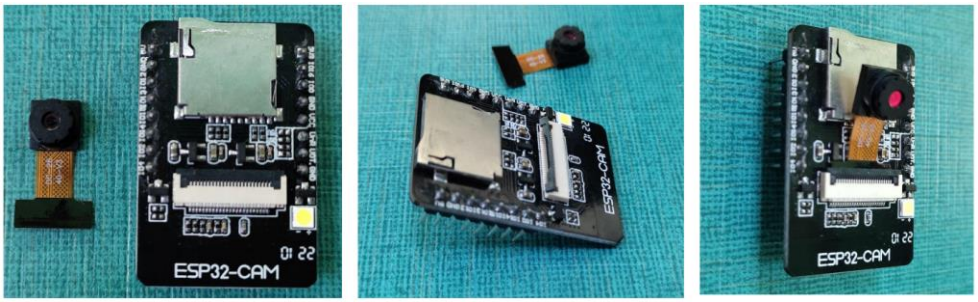
- ESP32 – CAM Module
- USB-To-TTL Converter
- Push Buttons
- LEDs
- Bread Board
- Jumper Wires

SOFTWARE REQUIREMENTS:

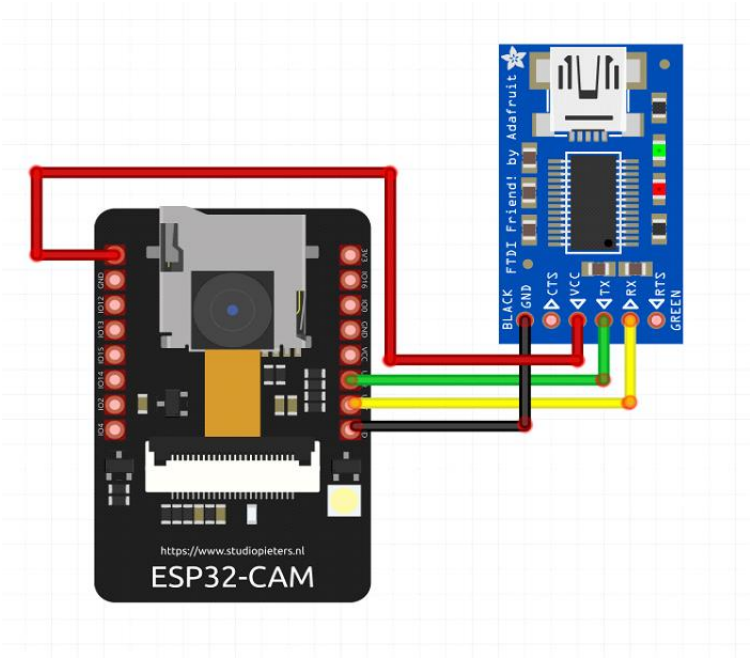
- Espressif Flash Download Tool
- Putty

WIRING ASSEMBLY:

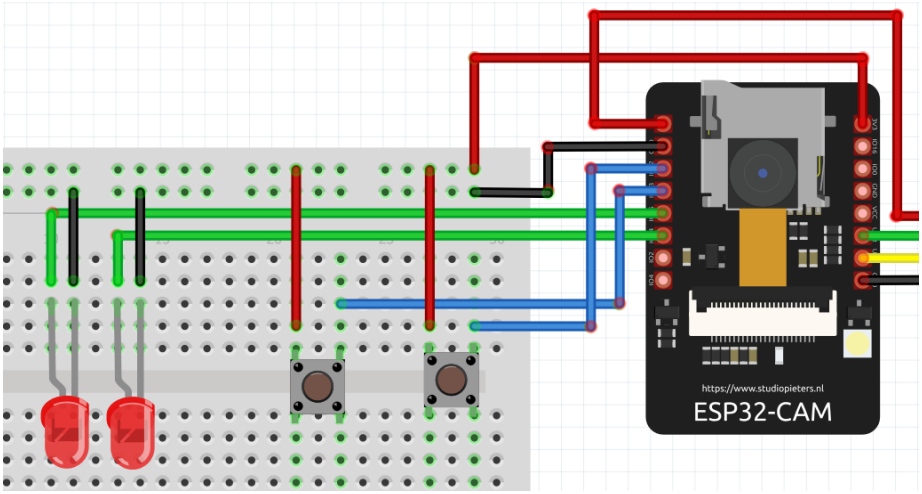
Firstly, connect the Camera Sensor with the FPC connector of ESP32-CAM, as shown:



Connect ESP32 - CAM with USB-to-TTL converter by using female-to-female jumper wires, as shown:

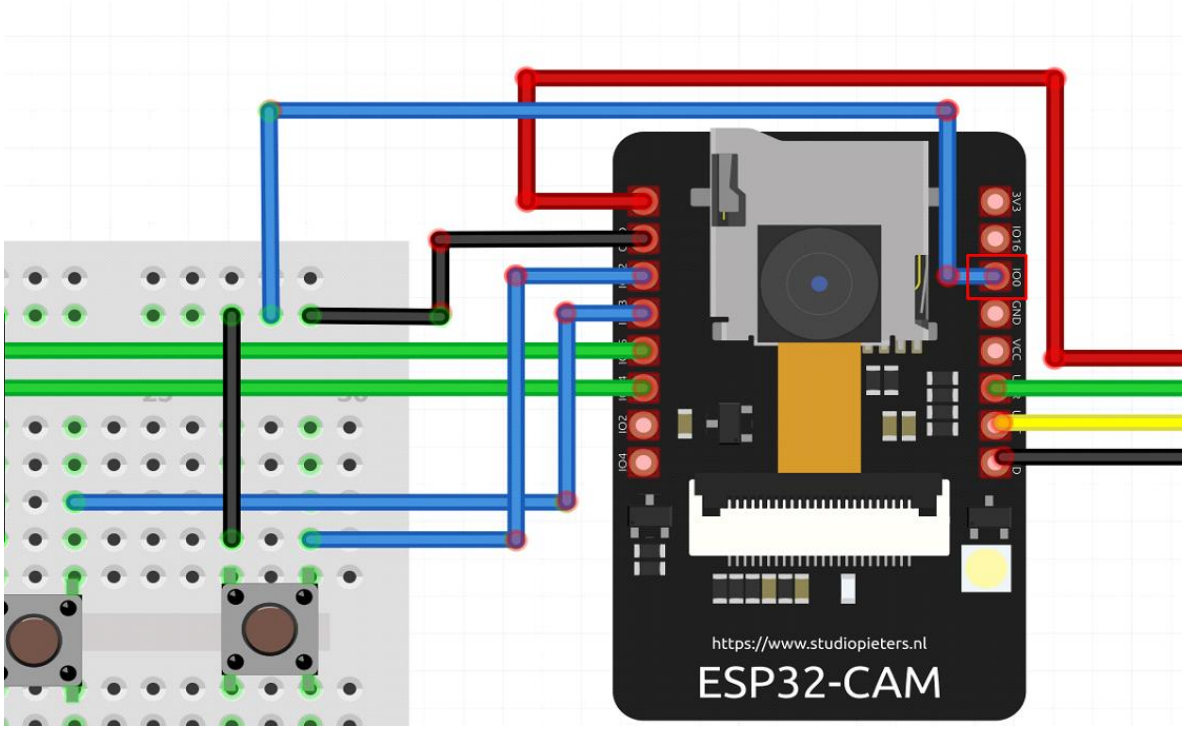


Connect Leds and Push Buttons with the above circuit, by using breadboard and jumper-wires.

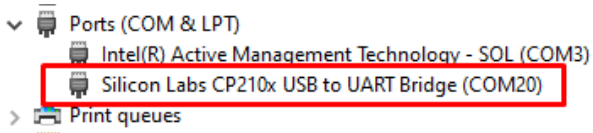


FIRMWARE UPLOAD & TESTING:

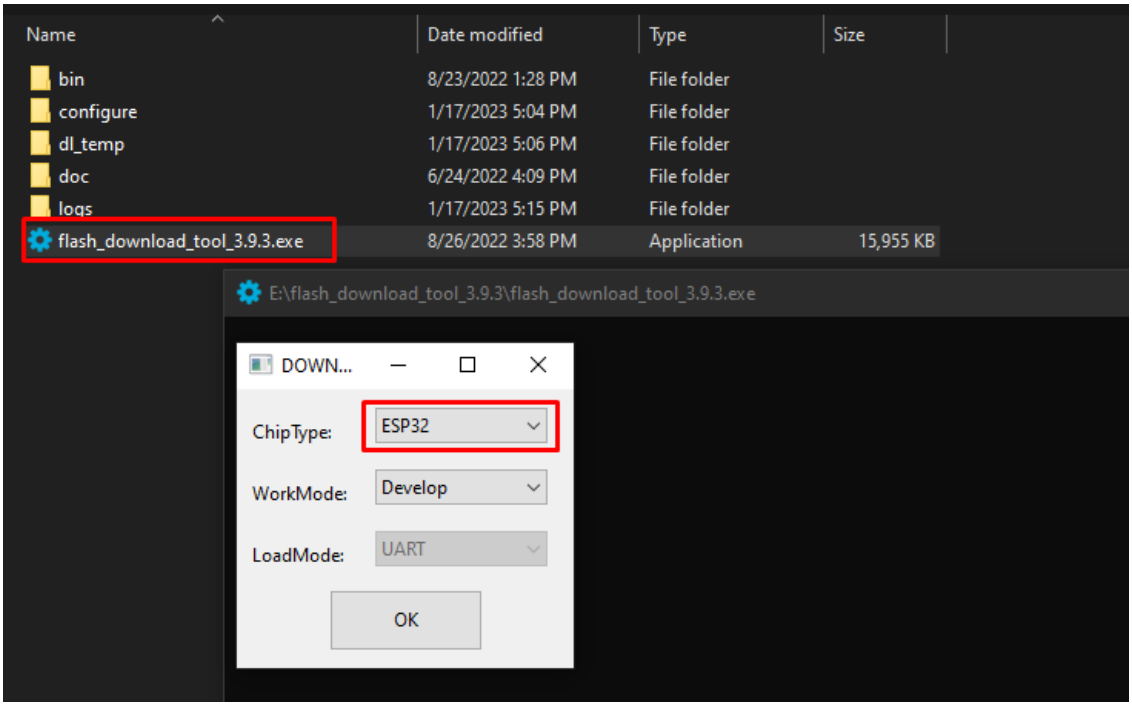
First you have to connect the **GPIO0** to **GND** and press the reset button of the ESP32-Cam board for "**FLASH MODE**".
Insert the USB-To-TTL Converter into the PC USB-Port.



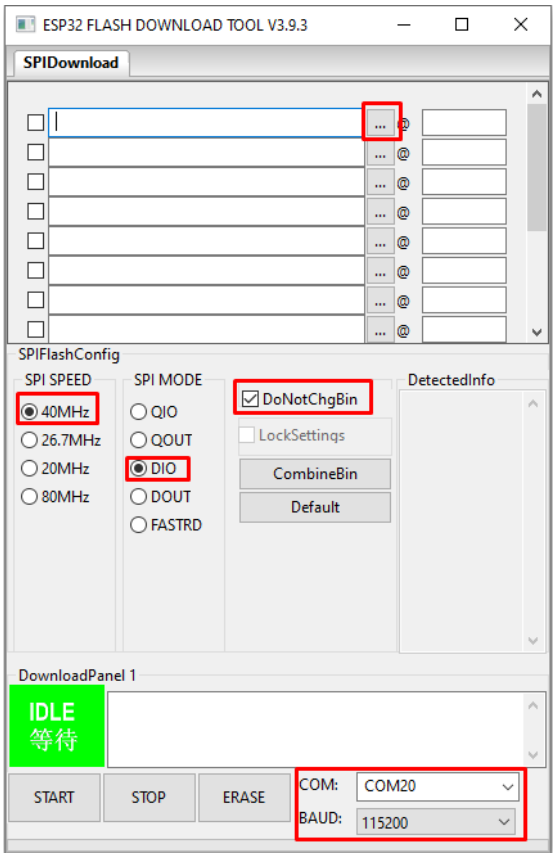
Open the **Device Manager->Ports** to check the COM Port.



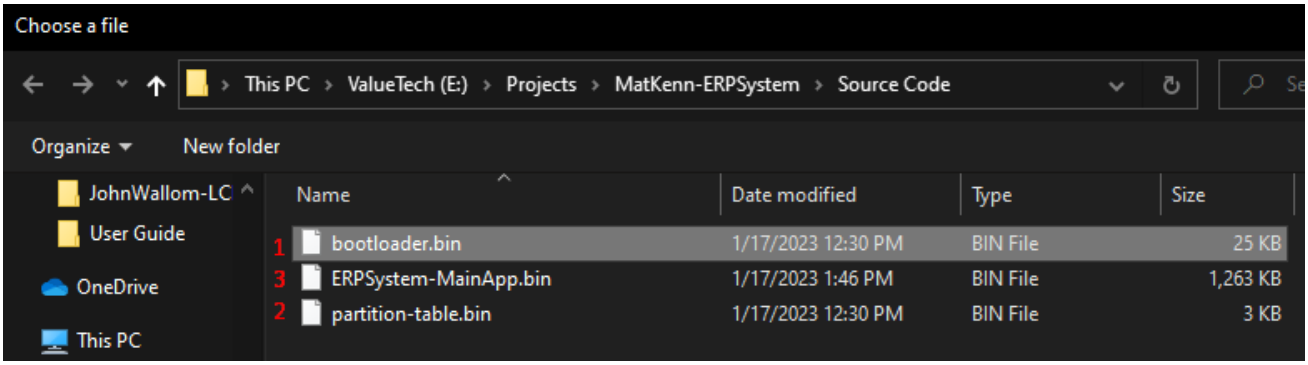
Now, open the “Flash Download Tool” and select “ESP32” as ChipType & press OK.

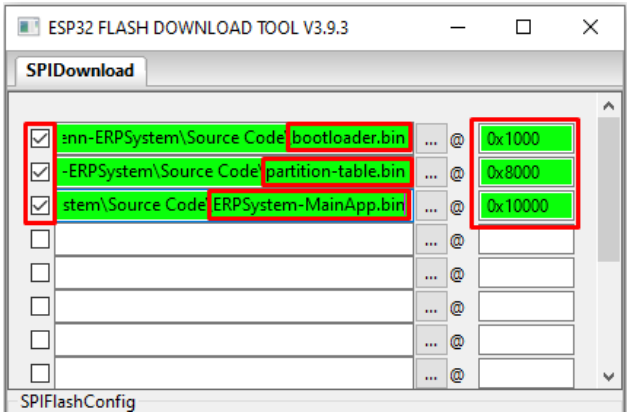


Now, Select the following configuration and the click on the “**...**” icon to select the bin files from the directories:

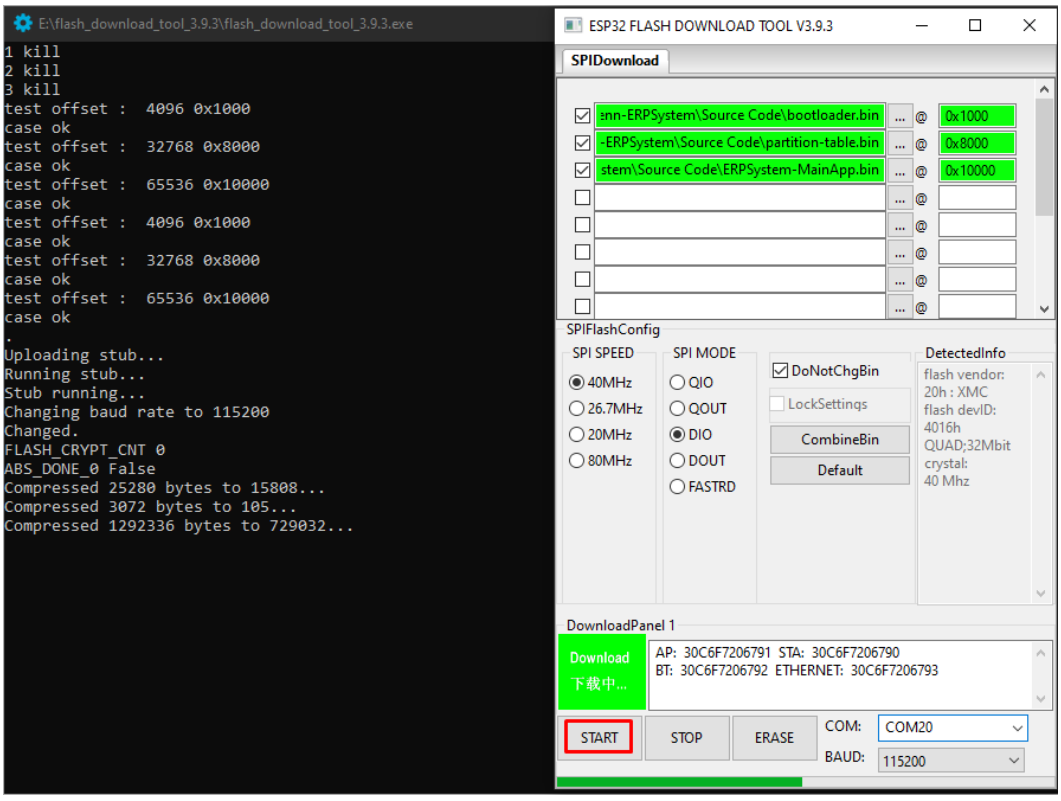


Select the “**bin files”** in the following order and insert the “**values”** as shown below.

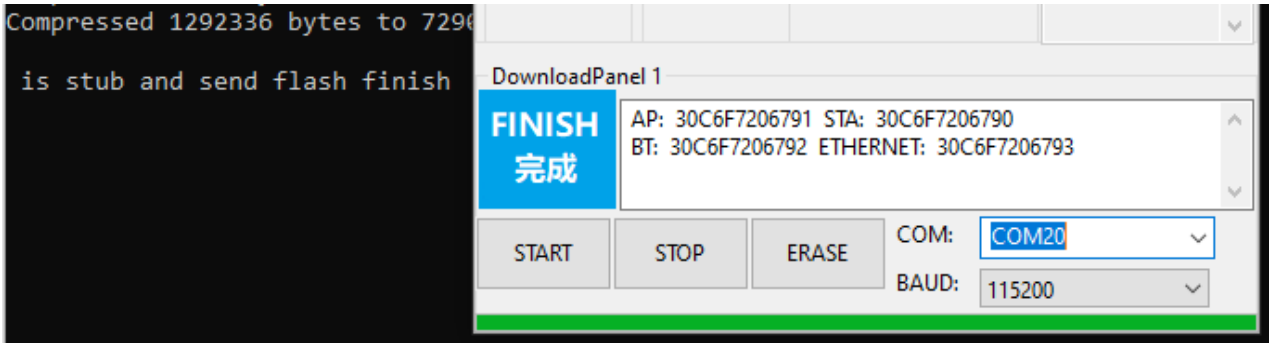




Click on the “**Start**” button & the firmware begins uploading.

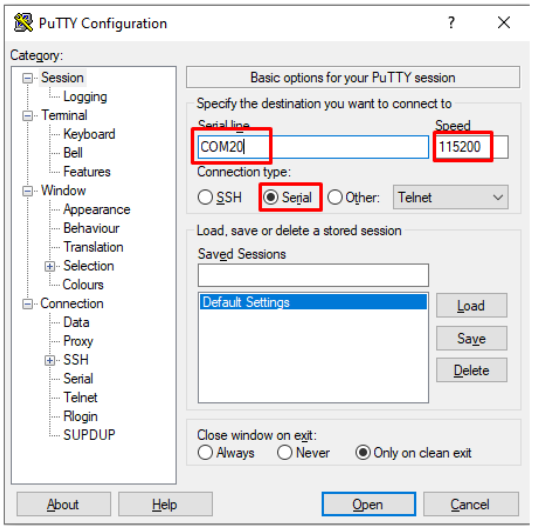


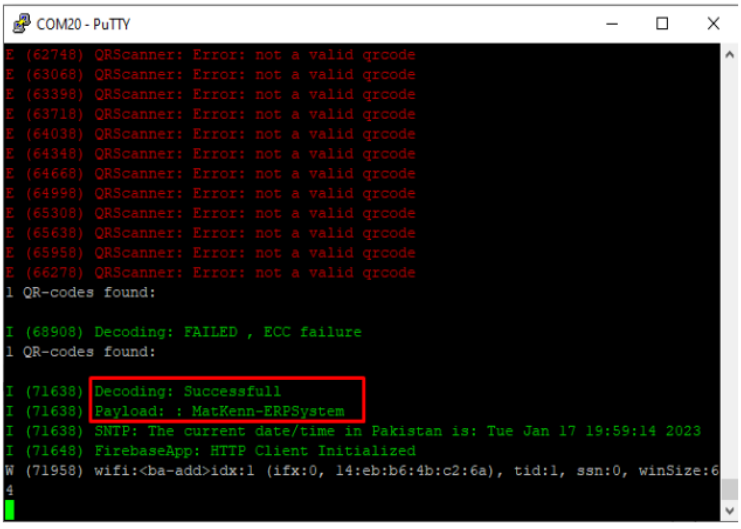
The following message shows that the firmware upload successfully!



Disconnect the **GPIO0** from **GND** and press the reset button, to select the "**RUN MODE**".

Open Putty for the output logs, as shown:
(Press the Reset Button if the Output Terminal is blank)





Open the official Firebase website and click on "**Get Started**".
Sign-in with the provided credentials and click on "**MatKenn-ERPSystem**".
Go to **Realtime Database->Data**, and here you can access the QR Scanner data.

